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High-frequency module, communication device, and method for manufacturing plurality of high-frequency modules

Abstract

A high-frequency module includes: a module board that has main surfaces which are opposed to each other; a plurality of high-frequency components that are arranged on at least one of the main surfaces; a resin member that covers at least one of the main surfaces and has a plurality of side surfaces along an outer edge of the module board; and a shield electrode layer that covers some of the side surfaces without covering other side surfaces.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This is a continuation of International Application No. PCT/JP2021/024570 filed on Jun. 29, 2021 which claims priority from Japanese Patent Application No. 2020-143576 filed on Aug. 27, 2020. The contents of these applications are incorporated herein by reference in their entireties.

BACKGROUND ART
Technical Field
(1) The present disclosure relates to a high-frequency module, a method for manufacturing a high-frequency module, and a communication device provided with a high-frequency module.
(2) In mobile communication devices such as mobile phones, an arrangement configuration of high-frequency components constituting a high-frequency front-end module has become complicated, especially with the development of multiband. Patent Document 1 discloses a front-end module in which a power amplifier, a switch, a filter, and the like are packaged. Patent Document 1: U.S. Patent Application Publication No. 2015/0133067
BRIEF SUMMARY
(3) In such a conventional front-end module, there is a concern that isolation between high-frequency components may degrade due to magnetic field coupling, electric field coupling, electromagnetic field coupling, or the like between the high-frequency components.
(4) Hence, the present disclosure provides a high-frequency module, a communication device, and a method for manufacturing a plurality of high-frequency modules that can suppress degradation of isolation between high-frequency components caused by magnetic field coupling, electric field coupling, electromagnetic field coupling, or the like.
(5) A high-frequency module according to one aspect of the present disclosure includes: a module board that has a first main surface and a second main surface which are opposed to each other; a plurality of high-frequency components that are arranged on at least one of the first main surface and the second main surface; a resin member that covers at least one of the first main surface and the second main surface and has a plurality of side surfaces, which include a first side surface and a second side surface, along an outer edge of the module board; and a shield electrode layer that covers the second side surface without necessarily covering the first side surface.
(6) According to the high-frequency module of the one aspect of the present disclosure, degradation of isolation between high-frequency components caused by magnetic field coupling, electric field coupling, electromagnetic field coupling, or the like can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a circuit configuration diagram of a high-frequency module and a communication device according to a first embodiment.
(2) FIG. **2** is a plan view of the high-frequency module according to the first embodiment.
(3) FIG. **3** is a sectional view of the high-frequency module according to the first embodiment.
(4) FIG. **4** is a sectional view of the high-frequency module according to the first embodiment.
(5) FIGS. **5**A-**5**D are diagrams illustrating a method for manufacturing the high-frequency module according to the first embodiment.
(6) FIG. **6** is a plan view of a high-frequency module according to a second embodiment.

(7) FIG. **7** is a plan view of a high-frequency module according to a third embodiment.
(8) FIG. **8** is a plan view of a high-frequency module according to a fourth embodiment.
(9) FIG. **9** is a transparent view of a collective board for the high-frequency modules according to the third and fourth embodiments.
(10) FIG. **10** is a plan view of a high-frequency module according to another embodiment.

DETAILED DESCRIPTION

(11) Embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings. Note that all of the embodiments described below are generic or specific examples. Numerical values, shapes, materials, components, arrangement and connection forms of the components, and the like shown in the following embodiments are examples, and are not intended to limit the present disclosure.
(12) Note that each drawing is a schematic diagram with emphasis, omission, or proportion adjustment performed as appropriate to illustrate the present disclosure. Thus, each drawing is not necessarily a strict illustration and may differ from the actual shapes, positioning, and proportions. In each drawing, the same reference characters are applied to substantially identical configurations and redundant description may be omitted or simplified.
(13) In each drawing below, an x axis and a y axis are axes that are orthogonal to each other on a plane which is parallel to a main surface of a module board. Specifically, when a module board has a rectangular shape in plan view, the x axis is parallel to a first side of the module board and the y axis is parallel to a second side, being orthogonal to the first side, of the module board. In addition, a z axis is an axis orthogonal to the main surface of the module board, and the positive direction of the z axis indicates the upward direction and the negative direction of the z axis indicates the downward direction.
(14) In the circuit configuration of the present disclosure, “connected” includes not only direct connection by connection terminals and/or wiring conductors, but also electrical connection via other circuit elements. Further, “connected between A and B” means being connected with both of A and B between A and B.
(15) In the component arrangement of the present disclosure, “plan view” means viewing an object by orthographic projection from the positive side of the z axis onto the xy plane. Further, a “distance between A and B” means a length of a line segment connecting a representative point in A and a representative point in B. Here, examples of a point used as a representative point can include a center point of an object or a point closest to a counterpart object, but the representative point is not limited to this. In addition, terms indicating relationships between elements, such as “parallel” and “orthogonal”, terms indicating the shape of elements, such as “rectangular”, and numerical ranges do not express strict meaning only but also include meaning of substantially equivalent ranges, for example, errors of approximately a few percent.
(16) In addition, “a component is arranged on a board” means that the component is arranged on the board while being in contact with the board, and that the component is arranged above the board without necessarily being in contact with the board (for example, the component is stacked on top of another component arranged on the board), and that the component is arranged in a manner to be partially or entirely embedded within the board. Further, “a component is arranged on a main surface of a board” means that the component is arranged on the main surface of the board while being in contact with the main surface, and that the component is arranged above the main surface without necessarily being in contact with the main surface, and that the component is arranged in a manner to be partially embedded within the board from the main surface side.

First Embodiment

(17) [1.1 Circuit Configurations of High-Frequency Module **1** and Communication Device **5**]
(18) Circuit configurations of a high-frequency module **1** and a communication device **5** according to the present embodiment will be described with reference to FIG. **1**. FIG. **1** is a circuit configuration diagram of the high-frequency module **1** and the communication device **5** according

to a first embodiment.

(19) [1.1.1 Circuit Configuration of Communication Device **5**]

(20) The circuit configuration of the communication device **5** will be first described. As illustrated in FIG. **1**, the communication device **5** according to the present embodiment includes the high-frequency module **1**, an antenna **2**, an RFIC **3**, and a BBIC **4**.

(21) The high-frequency module **1** transmits a high-frequency signal between the antenna **2** and the RFIC **3**. The internal configuration of the high-frequency module **1** will be described later.

(22) The antenna **2** is connected with an antenna connection terminal **101** of the high-frequency module **1**. The antenna **2** transmits a high-frequency signal outputted from the high-frequency module **1**, and also receives a high-frequency signal from the outside and outputs the high-frequency signal to the high-frequency module **1**.

(23) The RFIC **3** is an example of a signal processing circuit that processes a high-frequency signal. Specifically, the RFIC **3** processes a high-frequency reception signal, inputted via a reception path of the high-frequency module **1**, by down-conversion or the like and outputs the reception signal generated by this signal processing to the BBIC **4**. In addition, the RFIC **3** processes a transmission signal, inputted from the BBIC **4**, by up-conversion or the like and outputs the high-frequency transmission signal generated by this signal processing to a transmission path of the high-frequency module **1**. Further, the RFIC **3** includes a control unit that controls switches, amplifiers, and the like included in the high-frequency module **1**. Here, part or all of the function as the control unit of the RFIC **3** may be mounted on the outside of the RFIC **3**, such as on the BBIC **4** or the high-frequency module **1**.

(24) The BBIC **4** is a baseband signal processing circuit that processes a signal by using an intermediate frequency band which is lower in frequency than the high-frequency signal transmitted by the high-frequency module **1**. Examples of a signal processed in the BBIC **4** include an image signal for displaying an image and/or an audio signal for calls through speakers.

(25) In the communication device **5** according to the present embodiment, the antenna **2** and the BBIC **4** are optional components. That is, the communication device **5** does not have to include the antenna **2** and the BBIC **4**.

(26) [1.1.2 Circuit Configuration of High-Frequency Module **1**]

(27) The circuit configuration of the high-frequency module **1** will now be described. As illustrated in FIG. **1**, the high-frequency module **1** includes a power amplifier **11**, a low noise amplifier **21**, switches **51** to **53**, a duplexer **61**, a transmission-reception filter **62**, matching circuits (MN) **71** to **75**, the antenna connection terminal **101**, a high-frequency input terminal **111**, and a high-frequency output terminal **121**.

(28) The antenna connection terminal **101** is connected to the antenna **2**.

(29) The high-frequency input terminal **111** is a terminal for receiving a high-frequency transmission signal from the outside of the high-frequency module **1**. In the present embodiment, the high-frequency input terminal **111** is a terminal for receiving transmission signals of communication bands A and B from the RFIC **3**.

(30) The high-frequency output terminal **121** is a terminal for providing a high-frequency reception signal to the outside of the high-frequency module **1**. In the present embodiment, the high-frequency output terminal **121** is a terminal for providing reception signals of the communication bands A and B to the RFIC **3**.

(31) Here, the communication band means a frequency band predefined by a standards body or the like (for example, 3rd Generation Partnership Project (3GPP) and Institute of Electrical and Electronics Engineers (IEEE)) for communication systems.

(32) Here, the communication systems mean communication systems built by using the radio access technology (RAT). Examples of the communication systems include 5th generation new radio (SGNR) systems, long term evolution (LTE) systems, and wireless local area network (WLAN) systems, but the communication systems are not limited to these.

(33) A frequency division duplex (FDD) communication band is used as the communication band A. For example, bands **1**, **3**, **25**, and **66** for LTE and/or SGNR and the like can be used as the communication band A, but the communication band A is not limited to these.
(34) A time division duplex (TDD) communication band is used as the communication band B. For example, bands **34** and **39** for LTE and/or SGNR and the like can be used as the communication band B, but the communication band B is not limited to these.
(35) The power amplifier **11** is capable of amplifying transmission signals of the communication bands A and B received at the high-frequency input terminal **111**. An input of the power amplifier **11** is connected with the high-frequency input terminal **111** and an output of the power amplifier **11** is connected with the switch **52** via the matching circuit **72**.
(36) The low noise amplifier **21** is capable of amplifying reception signals of the communication bands A and B. An input of the low noise amplifier is connected with the switch **53** via the matching circuit **71** and an output of the low noise amplifier **21** is connected with the high-frequency output terminal **121**.
(37) The configuration of each of the power amplifier **11** and the low noise amplifier **21** is not especially limited. For example, the power amplifier **11** and/or the low noise amplifier **21** may have a single-stage configuration or a multistage configuration. Further, for example, the power amplifier **11** and/or the low noise amplifier **21** may be a differential amplifier which converts a high-frequency signal into a differential signal (that is, a complementary signal) and amplifies the differential signal.
(38) The duplexer **61** allows a high-frequency signal of the communication band A to pass through. The duplexer **61** transmits a transmission signal and a reception signal of the communication band A by the FDD method. The duplexer **61** includes a transmission filter **61**T and a reception filter **61**R.
(39) The transmission filter **61**T (A-Tx) is an example of a second filter and a third filter, and has a pass band including an uplink operating band of the communication band A. One end of the transmission filter **61**T is connected with the antenna connection terminal **101** via the matching circuit **73**, the switch **51**, and the matching circuit **75**. The other end of the transmission filter **61**T is connected with the output of the power amplifier **11** via the switch **52**.
(40) The uplink operating band means part of a communication band that is designated for uplink. In the high-frequency module **1**, the uplink operating band means a transmission band.
(41) The reception filter **61**R (A-Rx) is an example of a first filter and the third filter, and has a pass band including a downlink operating band of the communication band A. One end of the reception filter **61**R is connected with the antenna connection terminal **101** via the matching circuit **73**, the switch **51**, and the matching circuit **75**. The other end of the reception filter **61**R is connected with an input of the low noise amplifier **21** via the switch **53**.
(42) The downlink operating band means part of a communication band that is designated for downlink. In the high-frequency module **1**, the downlink operating band means a reception band.
(43) The transmission-reception filter **62** (B-TRx) is an example of the first filter, the second filter, and the third filter, and has a pass band including the communication band B. One end of the transmission-reception filter **62** is connected with the antenna connection terminal **101** via the matching circuit **74**, the switch **51**, and the matching circuit **75**. The other end of the transmission-reception filter **62** is connected with the output of the power amplifier **11** via the switch **52** and the matching circuit **72** and is connected with the input of the low noise amplifier **21** via the switches **52** and **53** and the matching circuit **71**.
(44) The switch **51** is connected between the antenna connection terminal **101** and each of the duplexer **61** and the transmission-reception filter **62**. The switch **51** includes terminals **511** to **513**. The terminal **511** is connected to the antenna connection terminal **101** via the matching circuit **75**. The terminal **512** is connected to the duplexer **61** via the matching circuit **73**. The terminal **513** is connected to the transmission-reception filter **62** via the matching circuit **74**.

(45) In this connection configuration, the switch **51** is capable of connecting the terminal **511** to at least one of the terminals **512** and **513** in response to, for example, a control signal from the RFIC **3**. That is, the switch **51** is capable of switching connection and disconnection between the antenna **2** and the duplexer **61** and capable of switching connection and disconnection between the antenna **2** and the transmission-reception filter **62**. The switch **51** is composed of, for example, a multi-connection switching circuit and is sometimes called an antenna switch.
(46) The switch **52** is connected between each of the transmission filter **61**T and the transmission-reception filter **62** and each of the power amplifier **11** and the low noise amplifier **21**. The switch **52** includes terminals **521** to **524**. The terminals **521** and **522** are connected with the transmission filter **61**T and the transmission-reception filter **62**, respectively. The terminal **523** is connected with the output of the power amplifier **11** via the matching circuit **72**. The terminal **524** is connected with a terminal **532** of the switch **53** and is connected with the input of the low noise amplifier **21** via the switch **53** and the matching circuit **71**.
(47) In this connection configuration, the switch **52** is capable of connecting the terminal **521** to the terminal **523** and capable of connecting the terminal **522** to either the terminal **523** or **524** in response to, for example, a control signal from the RFIC **3**. That is, the switch **52** is capable of switching connection and disconnection between the transmission filter **61**T and the power amplifier **11** and capable of connecting the transmission-reception filter **62** to either the power amplifier **11** or the low noise amplifier **21**. The switch **52** is composed of, for example, a multi-connection switching circuit.
(48) The switch **53** is connected between each of the reception filter **61**R and transmission-reception filter **62** and the low noise amplifier **21**. The switch **53** includes terminals **531** to **533**. The terminal **531** is connected with the input of the low noise amplifier **21** via the matching circuit **71**. The terminal **532** is connected with the terminal **524** of the switch **52** and is connected with the transmission-reception filter **62** via the switch **52**. The terminal **533** is connected with the reception filter **61**R.
(49) In this connection configuration, the switch **53** is capable of connecting the terminal **532** and/or the terminal **533** to the terminal **531** in response to, for example, a control signal from the RFIC **3**. That is, the switch **53** is capable of switching connection and disconnection between the reception filter **61**R and the low noise amplifier **21** and capable of switching connection and disconnection between the transmission-reception filter **62** and the low noise amplifier **21**. The switch **53** is composed of, for example, a multi-connection switching circuit.
(50) The matching circuit **71** is connected between each of the reception filter **61**R and transmission-reception filter **62** and the low noise amplifier **21**. Specifically, the matching circuit **71** is connected between the switch **53** and the input of the low noise amplifier **21**. The matching circuit **71** can achieve impedance matching between the low noise amplifier **21** and each of the reception filter **61**R and transmission-reception filter **62**. The matching circuit **71** includes an inductor **71**L and may further include a capacitor.
(51) The matching circuit **72** is connected between each of the transmission filter **61**T and transmission-reception filter **62** and the power amplifier **11**. Specifically, the matching circuit **72** is connected between the switch **52** and the input of the power amplifier **11**. The matching circuit **72** can achieve impedance matching between the power amplifier **11** and each of the transmission filter **61**T and transmission-reception filter **62**. The matching circuit **72** includes an inductor **72**L and may further include a capacitor.
(52) The matching circuit **73** is connected between the duplexer **61** and the antenna connection terminal **101**. Specifically, the matching circuit **73** is connected between the duplexer **61** and the switch **51**. The matching circuit **73** can achieve impedance matching between the duplexer **61** and the antenna **2**. The matching circuit **73** includes an inductor **73**L and may further include a capacitor.
(53) The matching circuit **74** is connected between the transmission-reception filter **62** and the

antenna connection terminal **101**. Specifically, the matching circuit **74** is connected between the transmission-reception filter **62** and the switch **51**. The matching circuit **74** can achieve impedance matching between the transmission-reception filter **62** and the antenna **2**. The matching circuit **74** includes an inductor **74**L and may further include a capacitor.

(54) The matching circuit **75** is connected between each of the duplexer **61** and transmission-reception filter **62** and the antenna connection terminal **101**. Specifically, the matching circuit **75** is connected between the switch **51** and the antenna connection terminal **101**. The matching circuit **75** can achieve impedance matching between each of the duplexer **61** and transmission-reception filter **62** and the antenna **2**. The matching circuit **75** includes an inductor **75**L and may further include a capacitor.

(55) Note that some of the circuit elements illustrated in FIG. **1** do not have to be included in the high-frequency module **1**. For example, the high-frequency module **1** may include at least two of the power amplifier **11**, the low noise amplifier **21**, the switches **51** to **53**, the duplexer **61**, the transmission-reception filter **62**, and the inductors **71**L to **75**L.

(56) [1.2 Component Arrangement in High-Frequency Module **1**]

(57) The component arrangement in the high-frequency module **1**, which is configured as described above, will now be specifically described with reference to FIGS. **2** to **4**.

(58) FIG. **2** is a plan view of the high-frequency module **1** according to the first embodiment. Specifically, FIG. **2** is a drawing illustrating a main surface **91***a* of a module board **91** viewed from the positive side of the z axis. Each of FIG. **3** and FIG. **4** is a sectional view of the high-frequency module **1** according to the first embodiment. Cross sections of the high-frequency module **1** in FIG. **3** and FIG. **4** are cross sections taken along a iii-iii line and a iv-iv line in FIG. **2**, respectively.

(59) As illustrated in FIGS. **2** to **4**, the high-frequency module **1** further includes the module board **91**, a resin member **92**, a shield electrode layer **95**, and a plurality of external connection terminals **150** in addition to a plurality of high-frequency components including the plurality of high-frequency elements illustrated in FIG. **1**. Here, FIG. **2** omits illustration of the resin member **92** and a top surface of the shield electrode layer **95**.

(60) The module board **91** has main surfaces **91***a* and **91***b* that are opposed to each other. The module board **91** has a rectangular shape in plan view in the present embodiment, but the shape of the module board **91** is not limited to the rectangular shape. Examples of a board used as the module board **91** include a low temperature co-fired ceramics (LTCC) board having a laminated structure of a plurality of dielectric layers, a high temperature co-fired ceramics (HTCC) board, a component-embedded board, a board having a redistribution layer (RDL), and a printed board, but the module board **91** is not limited to these.

(61) The main surface **91***a* is an example of a first main surface. A plurality of high-frequency components and the resin member **92** are arranged on the main surface **91***a*. In this example, the plurality of high-frequency components include the power amplifier **11**, the low noise amplifier **21**, the switches **51** to **53**, the duplexer **61**, the transmission-reception filter **62**, and the inductors **71**L to **75**L.

(62) The main surface **91***b* is an example of a second main surface. The plurality of external connection terminals **150** are arranged on the main surface **91***b*.

(63) The resin member **92** is arranged on the main surface **91***a* of the module board **91** and covers the main surface **91***a* and components arranged on the main surface **91***a*. The resin member **92** has a function of ensuring reliability such as mechanical strength and moisture resistance of the components on the main surface **91***a*.

(64) The resin member **92** is molded into a rectangular parallelepiped shape on the main surface **91***a*. Specifically, the resin member **92** has four side surfaces **921** to **924** along an outer edge (that is, four sides) of the module board **91** and has a top surface **925** above the main surface **91***a* of the module board **91**.

(65) The side surfaces **921** and **923** are examples of a first side surface and a third side surface. The

side surfaces **921** and **923** are along shorter sides of the module board **91** and are opposed to each other. The side surfaces **922** and **924** are examples of a second side surface and a fourth side surface. The side surfaces **922** and **924** are along longer sides of the module board **91** and are opposed to each other.

(66) The shield electrode layer **95** is a metal film that is formed by sputtering or the like, for example, and covers the top surface **925** and the side surfaces **922** and **924** of the resin member **92**. The shield electrode layer **95** is set to ground potential and suppresses external noise from entering the components constituting the high-frequency module **1**.

(67) Specifically, the shield electrode layer **95** covers the side surfaces **922** and **924** without necessarily covering the side surfaces **921** and **923** among the four side surfaces **921** to **924** of the resin member **92**. In other words, the shield electrode layer **95** covers only the two side surfaces **922** and **924** that are opposed to each other, among the four side surfaces **921** to **924** of the resin member **92**.

(68) The low noise amplifier **21** and the switch **53** are incorporated in one semiconductor integrated circuit **20**. The semiconductor integrated circuit **20** is an electronic component that includes electronic circuits formed on a surface or in the inside of a semiconductor chip (also called die). The semiconductor integrated circuit **20** has a rectangular shape in plan view, in the present embodiment. However, the shape of the semiconductor integrated circuit **20** is not limited to the rectangular shape. In addition, circuit elements incorporated in the semiconductor integrated circuit **20** are not limited to the low noise amplifier **21** and the switch **53**. For example, the switch **51** may be incorporated in the semiconductor integrated circuit **20**.

(69) The semiconductor integrated circuit **20** may be, for example, made of complementary metal oxide semiconductor (CMOS) and specifically, may be formed by a silicon on insulator (SOI) process. The semiconductor integrated circuit **20** can be accordingly manufactured inexpensively. Here, the semiconductor integrated circuit **20** may be made of at least one of GaAs, SiGe, and GaN. The semiconductor integrated circuit **20** with high quality can be thus realized.

(70) Each of the transmission filter **61**T, the reception filter **61**R, and the transmission-reception filter **62** may be any of a surface acoustic wave filter, an acoustic wave filter using a bulk acoustic wave (BAW), an LC resonance filter, and a dielectric filter, for example, and further, each of the transmission filter **61**T, the reception filter **61**R, and the transmission-reception filter **62** is not limited to these.

(71) Each of the inductors **71**L to **75**L is composed of a coil and is mounted as, for example, a surface mount device (SMD). Here, the inductors **71**L to **75**L do not have to be SMDs but may be formed inside the module board **91**.

(72) The inductor **71**L is an example of a first inductor and is arranged near the side surface **921** of the resin member **92**. Specifically, the inductor **71**L is closer to the side surface **921** of the side surfaces **921** and **922**. Further, the inductor **71**L is closer to the side surface **921** than the low noise amplifier **21**.

(73) A winding axis **71**La of a coil constituting the inductor **71**L is parallel to the side surfaces **922** and **924** and is orthogonal to the side surfaces **921** and **923**. That is, the winding axis **71**La intersects with the side surfaces **921** and **923** but does not intersect with the side surfaces **922** and **924**. The winding axis **71**La accordingly does not intersect with the shield electrode layer **95**.

(74) The inductor **72**L is an example of a second inductor and is arranged near the side surface **923** of the resin member **92**. Specifically, the inductor **72**L is closer to the side surface **923** of the side surfaces **923** and **922**. Further, the inductor **72**L is closer to the side surface **923** than the power amplifier **11**.

(75) A winding axis **72**La of a coil constituting the inductor **72**L is parallel to the side surfaces **922** and **924** and is orthogonal to the side surfaces **921** and **923**. That is, the winding axis **72**La intersects with the side surfaces **921** and **923** but does not intersect with the side surfaces **922** and **924**. The winding axis **72**La accordingly does not intersect with the shield electrode layer **95**.

(76) The inductor **73**L is arranged near the side surface **924** of the resin member **92**. Here, a winding axis **73**La of a coil constituting the inductor **73**L does not intersect with the side surfaces **921** and **923** of the resin member **92** but intersects with the side surfaces **922** and **924** of the resin member **92**.
(77) The inductor **74**L is an example of a third inductor and is arranged near the center of the module board **91**. A winding axis **74**La of a coil constituting the inductor **74**L is parallel to the side surfaces **922** and **924** and is orthogonal to the side surfaces **921** and **923**. That is, the winding axis **74**La intersects with the side surfaces **921** and **923** but does not intersect with the side surfaces **922** and **924**. The winding axis **74**La accordingly does not intersect with the shield electrode layer **95**.
(78) The inductor **75**L is an example of the third inductor and is arranged near the side surface **921** of the resin member **92**. Specifically, the inductor **75**L is closer to the side surface **921** of the side surfaces **921** and **924**. Further, the inductor **75**L is closer to the side surface **921** than the duplexer **61** and the transmission-reception filter **62**.
(79) A winding axis **75**La of a coil constituting the inductor **75**L is parallel to the side surfaces **922** and **924** and is orthogonal to the side surfaces **921** and **923**. That is, the winding axis **75**La intersects with the side surfaces **921** and **923** but does not intersect with the side surfaces **922** and **924**. The winding axis **75**La accordingly does not intersect with the shield electrode layer **95**.
(80) The plurality of external connection terminals **150** include the antenna connection terminal **101**, the high-frequency input terminal **111**, and the high-frequency output terminal **121** which are illustrated in FIG. **1**. Each of the plurality of external connection terminals **150** is connected with, for example, an input/output terminal and/or a ground terminal on a mother board arranged in the negative direction of the z axis of the high-frequency module **1**. Pad electrodes can be used as the plurality of external connection terminals **150**, but the external connection terminal **150** is not limited to this.
(81) [1.3 Method for Manufacturing High-Frequency Module **1**]
(82) A method for manufacturing the high-frequency module **1** will now be described with reference to FIGS. **5**A-**5**D. FIGS. **5**A-**5**D are diagrams illustrating the method for manufacturing the high-frequency module **1** according to the first embodiment.
(83) (a) Preparation Process
(84) A collective board **910** on which a plurality of high-frequency components are mounted is first prepared. The collective board **910** includes a plurality of module boards **91**, thus being an aggregate of the plurality of module boards **91**. In FIGS. **5**A-**5**D, the plurality of module boards **91** are aligned in the collective board **910**.
(85) (b) Resin Molding Process
(86) Next, the resin member **92** is molded so as to cover the main surface of the collective board **910** and a plurality of high-frequency components on the main surface. For example, resin is poured into a mold placed on the collective board **910** and then the resin hardens to mold the resin member **92**. The surface of the resin member **92** which is molded is polished or ground as required.
(87) (c) Film Forming Process
(88) Next, a metal film is formed on the surface of the resin member **92**, which covers the collective board **910**, as the shield electrode layer **95**. That is, the shield electrode layers **95** of a plurality of high-frequency modules **1** are collectively formed. A forming method for the metal film is not especially limited but, for example, a sputtering method or the like can be employed.
(89) (d) Cutting Process
(90) The collective board **910**, on which the resin member **92** and the shield electrode layer **95** are formed, is cut and thus a plurality of high-frequency modules **1** are finally manufactured. As a result, the shield electrode layer **95** is not formed on cut surfaces of each of the high-frequency modules **1**. In FIGS. **5**A-**5**D, the shield electrode layer **95** is formed on two side surfaces **922** and **924** of the resin member **92** but the shield electrode layer **95** is not formed on two side surfaces **921** and **923** in each of the high-frequency modules **1**.

(91) [1.4 Effects, Etc.]

(92) As described above, the high-frequency module **1** according to the present embodiment includes: the module board **91** that has the main surfaces **91***a* and **91***b*, which are opposed to each other; a plurality of high-frequency components that are arranged on at least one of the main surfaces **91***a* and **91***b*; the resin member **92** that covers at least one of the main surfaces **91***a* and **91***b* and has a plurality of side surfaces, which include the side surface **921** and/or the side surface **923** and the side surface **922** and/or the side surface **924**, along the outer edge of the module board **91**; and the shield electrode layer **95** that covers the side surface **922** and/or the side surface **924** without necessarily covering the side surface **921** and/or the side surface **923**.

(93) According to this configuration, the side surface **922** and/or the side surface **924** can be covered by the shield electrode layer **95** without necessarily covering the side surface **921** and/or the side surface **923** by the shield electrode layer **95**. A high-frequency component may be magnetically coupled, electrically coupled, or electromagnetically coupled with other high-frequency components via the shield electrode layer **95**. Therefore, the configuration in which the side surface **921** and/or the side surface **923** among a plurality of side surfaces are/is not covered by the shield electrode layer **95** can suppress such coupling and improve isolation between the plurality of high-frequency components. In addition, the shield electrode layer **95** covers the side surface **922** and/or the side surface **924** among the plurality of side surfaces, being able to suppress external noise from entering the high-frequency components.

(94) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the reception filter **61**R, the low noise amplifier **21**, and the matching circuit **71** including the inductor **71**L, which is connected between the reception filter **61**R and the low noise amplifier **21**, and the inductor **71**L may be closer to the side surface **921** of the side surfaces **921** and **922**.

(95) According to this configuration, the inductor **71**L can be arranged closer to the side surface **921** which is not covered by the shield electrode layer **95** than to the side surface **922** which is covered by the shield electrode layer **95**. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95** can be effectively suppressed. Further, the inductor **71**L is arranged on a reception path. Accordingly, coupling between the inductor **71**L and other high-frequency components is suppressed, being able to suppress degradation of reception sensitivity.

(96) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the reception filter **61**R, the low noise amplifier **21**, and the matching circuit **71** including the inductor **71**L, which is connected between the reception filter **61**R and the low noise amplifier **21**, and the inductor **71**L may be closer to the side surface **921** than the low noise amplifier **21**.

(97) According to this configuration, the inductor **71**L can be arranged closer to the side surface **921**, which is not covered by the shield electrode layer **95**, than the low noise amplifier **21**. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95** can be effectively suppressed. Further, the inductor **71**L is arranged on a reception path. Accordingly, coupling between the inductor **71**L and other high-frequency components is suppressed, being able to suppress degradation of reception sensitivity.

(98) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the reception filter **61**R, the low noise amplifier **21**, and the matching circuit **71** including the inductor **71**L, which is connected between the reception filter **61**R and the low noise amplifier **21**, and the winding axis **71**La of a coil constituting the inductor **71**L may intersect with the side surface **921** but does not have to intersect with the side surface **922**.

(99) According to this configuration, the inductor **71**L can be arranged so that the winding axis

71La of a coil constituting the inductor **71**L does not intersect with the side surface **922** covered by the shield electrode layer **95**. In general, inductors generate a strong magnetic force along a winding axis of a coil and are prone to generate magnetic field coupling with components on the winding axis. Therefore, the configuration in which the winding axis **71**La does not intersect with the side surface **922** covered by the shield electrode layer **95** can effectively suppress magnetic field coupling of the inductor **71**L with other high-frequency components via the shield electrode layer **95**. Further, the inductor **71**L is arranged on a reception path. Accordingly, coupling between the inductor **71**L and other high-frequency components is suppressed, being able to suppress degradation of reception sensitivity.

(100) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the transmission filter **61**T, the power amplifier **11**, and the inductor **72**L, which is connected between the transmission filter **61**T and the power amplifier **11**, and the inductor **72**L may be closer to the side surface **923** of the side surfaces **922** and **923**.

(101) According to this configuration, the inductor **72**L can be arranged closer to the side surface **923** which is not covered by the shield electrode layer **95** than to the side surface **922** which is covered by the shield electrode layer **95**. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95** can be effectively suppressed. Further, the inductor **72**L is arranged on a transmission path. Accordingly, coupling between the inductor **72**L and other high-frequency components is suppressed, being able to suppress degradation of quality of a transmission signal.

(102) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the transmission filter **61**T, the power amplifier **11**, and the inductor **72**L, which is connected between the transmission filter **61**T and the power amplifier **11**, and the inductor **72**L may be closer to the side surface **923** than the power amplifier **11**.

(103) According to this configuration, the inductor **72**L can be arranged closer to the side surface **923** which is not covered by the shield electrode layer **95** than the power amplifier **11**. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95** can be effectively suppressed. Further, the inductor **72**L is arranged on a transmission path. Accordingly, coupling between the inductor **72**L and other high-frequency components is suppressed, being able to suppress degradation of quality of a transmission signal.

(104) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the transmission filter **61**T, the power amplifier **11**, and the inductor **72**L, which is connected between the transmission filter **61**T and the power amplifier **11**, and the winding axis **72**La of a coil constituting the inductor **72**L may intersect with the side surface **922** but does not have to intersect with the side surface **923**.

(105) According to this configuration, the inductor **72**L can be arranged so that the winding axis **72**La of a coil constituting the inductor **72**L does not intersect with the side surface **922** covered by the shield electrode layer **95**. In general, inductors generate a strong magnetic force along a winding axis of a coil and are prone to generate magnetic field coupling with components on the winding axis. Therefore, the configuration in which the winding axis **72**La does not intersect with the side surface **922** covered by the shield electrode layer **95** can effectively suppress magnetic field coupling of the inductor **72**L with other high-frequency components via the shield electrode layer **95**. Further, the inductor **72**L is arranged on a transmission path. Accordingly, coupling between the inductor **72**L and other high-frequency components is suppressed, being able to suppress degradation of quality of a transmission signal.

(106) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the transmission-reception filter **62** and

the inductor **75**L, which is connected between the transmission-reception filter **62** and the antenna connection terminal **101**, and the inductor **75**L may be closer to the side surface **921** of the side surfaces **921** and **924**.
(107) According to this configuration, the inductor **75**L can be arranged closer to the side surface **921** which is not covered by the shield electrode layer **95** than to the side surface **924** which is covered by the shield electrode layer **95**. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95** can be effectively suppressed. Further, the inductor **75**L is arranged on a transmission/reception path. Accordingly, coupling between the inductor **75**L and other high-frequency components is suppressed, being able to suppress degradation of a quality of a transmission signal and degradation of reception sensitivity.
(108) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the transmission-reception filter **62** and the inductor **75**L, which is connected between the transmission-reception filter **62** and the antenna connection terminal **101**, and the inductor **75**L may be closer to the side surface **921** than the transmission-reception filter **62**.
(109) According to this configuration, the inductor **75**L can be arranged closer to the side surface **921**, which is not covered by the shield electrode layer **95**, than the transmission-reception filter **62**. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95** can be effectively suppressed. Further, the inductor **75**L is arranged on a transmission/reception path. Accordingly, coupling between the inductor **75**L and other high-frequency components is suppressed, being able to suppress degradation of a quality of a transmission signal and degradation of reception sensitivity.
(110) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of high-frequency components may include the transmission-reception filter **62** and the inductor **74**L and/or the inductor **75**L, which are/is connected between the transmission-reception filter **62** and the antenna connection terminal **101**, and the winding axis **74**La of a coil constituting the inductor **74**L and/or the winding axis **75**La of a coil constituting the inductor **75**L may intersect with the side surface **921** but does not have to intersect with the side surface **924**.
(111) According to this configuration, the inductor **74**L and/or the inductor **75**L can be arranged so that the winding axis **74**La of a coil constituting the inductor **74**L and/or the winding axis **75**La of a coil constituting the inductor **75**L do/does not intersect with the side surface **924** covered by the shield electrode layer **95**. In general, inductors generate a strong magnetic force along a winding axis of a coil and are prone to generate magnetic field coupling with components on the winding axis. Therefore, the configuration in which the winding axis **74**La and/or the winding axis **75**La do/does not intersect with the side surface **924** covered by the shield electrode layer **95** can effectively suppress magnetic field coupling of the inductor **74**L and/or the inductor **75**L with other high-frequency components via the shield electrode layer **95**. Further, the inductor **75**L is arranged on a transmission/reception path. Accordingly, coupling between the inductor **75**L and other high-frequency components is suppressed, being able to suppress degradation of a quality of a transmission signal and degradation of reception sensitivity.
(112) Further, for example, in the high-frequency module **1** according to the present embodiment, the plurality of side surfaces may include the side surfaces **921** to **924** and the shield electrode layer **95** may cover the side surfaces **922** and **924** without necessarily covering the side surfaces **921** and **923**.
(113) According to this configuration, two side surfaces **921** and **923**, which are not covered by the shield electrode layer **95**, and two side surfaces **922** and **924**, which are covered by the shield electrode layer **95**, can be arranged. Accordingly, a balance can be achieved between suppression of coupling between high-frequency components and suppression of external noise intrusion.

(114) Further, for example, in the high-frequency module **1** according to the present embodiment, the side surfaces **921** and **923** may be opposed to each other and the side surfaces **922** and **924** may be opposed to each other.
(115) According to this configuration, the side surfaces **921** and **923**, which are not covered by the shield electrode layer **95**, can be made opposed to each other and the side surfaces **922** and **924**, which are covered by the shield electrode layer **95**, can be made opposed to each other. This facilitates arranging an inductor so that a winding axis does not intersect with a side surface covered by the shield electrode layer **95**, being able to easily arrange a plurality of high-frequency components so as to suppress coupling between the high-frequency components.
(116) Further, the communication device **5** according to the present embodiment includes the RFIC **3** that processes a high-frequency signal and the high-frequency module **1** that transmits a high-frequency signal between the RFIC **3** and the antenna **2**.
(117) According to this configuration, the communication device **5** can achieve the same effects as those of the high-frequency module **1**.
(118) Further, the method for manufacturing the high-frequency module **1** according to the present embodiment is a method for manufacturing a plurality of high-frequency modules **1** and includes: a resin molding process for molding the resin member **92** that covers the main surface of the collective board **910**, which includes a plurality of module boards **91** on which a plurality of high-frequency components are arranged; a film forming process for forming a metal film on the surface of the resin member **92**, which is molded, as the shield electrode layer **95**; and a cutting process for cutting the collective board **910** after the forming of the shield electrode layer **95** so as to manufacture a plurality of high-frequency modules **1**.
(119) According to the method, the shield electrode layer **95** can be collectively formed for the plurality of high-frequency modules **1** before cutting the collective board **910**. Thus, a plurality of high-frequency modules **1** can be more easily manufactured than a method in which the shield electrode layers **95** are separately formed on a plurality of high-frequency modules **1**. In addition, the shield electrode layer **95** is not formed on cut surfaces of a plurality of the high-frequency modules **1** thus manufactured. Accordingly, the plurality of the high-frequency modules **1** thus manufactured can suppress coupling between a plurality of high-frequency components via the shield electrode layer **95**.
Second Embodiment
(120) A second embodiment will now be described. In the present embodiment, a position of a side surface covered by a shield electrode layer is mainly different from that of the first embodiment described above. The present embodiment will be described below focusing on the difference from the above-described first embodiment with reference to the accompanying drawing.
(121) The circuit configuration of a high-frequency module **1**A according to the present embodiment is the same as that of the high-frequency module **1** according to the first embodiment and the illustration and description thereof are therefore omitted.
(122) [2.1 Component Arrangement in High-Frequency Module **1**A]
(123) The component arrangement in the high-frequency module **1**A according to the present embodiment will now be specifically described with reference to FIG. **6**. FIG. **6** is a plan view of the high-frequency module **1**A according to the second embodiment.
(124) In the present embodiment, the side surfaces **922** and **924** of the resin member **92** correspond to the first side surface and the third side surface, and the side surfaces **921** and **923** correspond to the second side surface and the fourth side surface.
(125) A shield electrode layer **95**A covers the side surfaces **921** and **923** without necessarily covering the side surfaces **922** and **924** among the four side surfaces **921** to **924** of the resin member **92**. In other words, the shield electrode layer **95**A covers only the two side surfaces **921** and **923** that are opposed to each other, among the four side surfaces **921** to **924** of the resin member **92**.

(126) The inductor **71**L is an example of the first inductor and is arranged near the side surface **922** of the resin member **92**. Specifically, the inductor **71**L is closer to the side surface **922** of the side surfaces **921** and **922**. Further, the inductor **71**L is closer to the side surface **922** than the low noise amplifier **21**.
(127) The winding axis **71**La of a coil constituting the inductor **71**L is parallel to the side surfaces **921** and **923** and is orthogonal to the side surfaces **922** and **924**. That is, the winding axis **71**La intersects with the side surfaces **922** and **924** but does not intersect with the side surfaces **921** and **923**. The winding axis **71**La accordingly does not intersect with the shield electrode layer **95**.
(128) The inductor **73**L is an example of the third inductor and is arranged near the side surface **924** of the resin member **92**. Specifically, the inductor **73**L is closer to the side surface **924** of the side surfaces **921** and **924**. Further, the inductor **73**L is closer to the side surface **924** than the duplexer **61** and the transmission-reception filter **62**.
(129) The winding axis **73**La of a coil constituting the inductor **73**L is parallel to the side surfaces **921** and **923** and is orthogonal to the side surfaces **922** and **924**. That is, the winding axis **73**La intersects with the side surfaces **922** and **924** but does not intersect with the side surfaces **921** and **923**. The winding axis **73**La accordingly does not intersect with the shield electrode layer **95**.
(130) [2.2 Effects, Etc.]
(131) As described above, the high-frequency module **1**A according to the present embodiment includes: the module board **91** that has the main surfaces **91***a* and **91***b*, which are opposed to each other; a plurality of high-frequency components that are arranged on at least one of the main surfaces **91***a* and **91***b*; the resin member **92** that covers at least one of the main surfaces **91***a* and **91***b* and has a plurality of side surfaces, which include the side surface **921** and/or the side surface **923** and the side surface **922** and/or the side surface **924**, along the outer edge of the module board **91**; and the shield electrode layer **95**A that covers the side surface **921** and/or the side surface **923** without necessarily covering the side surface **922** and/or the side surface **924**.
(132) According to this configuration, the side surface **921** and/or the side surface **923** can be covered by the shield electrode layer **95**A without necessarily covering the side surface **922** and/or the side surface **924** by the shield electrode layer **95**A. A high-frequency component may be magnetically coupled, electrically coupled, or electromagnetically coupled with other high-frequency components via the shield electrode layer **95**A. Therefore, the configuration in which the side surface **922** and/or the side surface **924** among a plurality of side surfaces are/is not covered by the shield electrode layer **95**A can suppress such coupling and improve isolation between the plurality of high-frequency components. In addition, the shield electrode layer **95**A covers the side surface **921** and/or the side surface **923** among the plurality of side surfaces, being able to suppress external noise from entering the high-frequency components.
(133) Further, for example, in the high-frequency module **1**A according to the present embodiment, the plurality of high-frequency components may include the reception filter **61**R, the low noise amplifier **21**, and the inductor **71**L, which is connected between the reception filter **61**R and the low noise amplifier **21**, and the inductor **71**L may be closer to the side surface **922** of the side surfaces **921** and **922**.
(134) According to this configuration, the inductor **71**L can be arranged closer to the side surface **922** which is not covered by the shield electrode layer **95**A than to the side surface **921** which is covered by the shield electrode layer **95**A. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95**A can be effectively suppressed. Further, the inductor **71**L is arranged on a reception path. Accordingly, coupling between the inductor **71**L and other high-frequency components is suppressed, being able to suppress degradation of reception sensitivity.
(135) Further, for example, in the high-frequency module **1**A according to the present embodiment, the plurality of high-frequency components may include the reception filter **61**R, the low noise amplifier **21**, and the inductor **71**L, which is connected between the reception filter **61**R and the

low noise amplifier **21**, and the inductor **71**L may be closer to the side surface **922** than the low noise amplifier **21**.
(136) According to this configuration, the inductor **71**L can be arranged closer to the side surface **922**, which is not covered by the shield electrode layer **95**A, than the low noise amplifier **21**. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95**A can be effectively suppressed. Further, the inductor **71**L is arranged on a reception path. Accordingly, coupling between the inductor **71**L and other high-frequency components is suppressed, being able to suppress degradation of reception sensitivity.
(137) Further, for example, in the high-frequency module **1**A according to the present embodiment, the plurality of high-frequency components may include the reception filter **61**R, the low noise amplifier **21**, and the inductor **71**L, which is connected between the reception filter **61**R and the low noise amplifier **21**, and the winding axis **71**La of a coil constituting the inductor **71**L may intersect with the side surface **922** but does not have to intersect with the side surface **921**.
(138) According to this configuration, the inductor **71**L can be arranged so that the winding axis **71**La of a coil constituting the inductor **71**L does not intersect with the side surface **921** covered by the shield electrode layer **95**A. In general, inductors generate a strong magnetic force along a winding axis of a coil and are prone to generate magnetic field coupling with components on the winding axis. Therefore, the configuration in which the winding axis **71**La does not intersect with the side surface **921** covered by the shield electrode layer **95**A can effectively suppress magnetic field coupling of the inductor **71**L with other high-frequency components via the shield electrode layer **95**A. Further, the inductor **71**L is arranged on a reception path. Accordingly, coupling between the inductor **71**L and other high-frequency components is suppressed, being able to suppress degradation of reception sensitivity.
(139) Further, for example, in the high-frequency module **1**A according to the present embodiment, the plurality of high-frequency components may include the duplexer **61** and the inductor **73**L, which is connected between the duplexer **61** and the antenna connection terminal **101**, and the inductor **73**L may be closer to the side surface **924** of the side surfaces **921** and **924**.
(140) According to this configuration, the inductor **73**L can be arranged closer to the side surface **924** which is not covered by the shield electrode layer **95**A than to the side surface **921** which is covered by the shield electrode layer **95**A. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95**A can be effectively suppressed. Further, the inductor **73**L is arranged on a transmission/reception path in the communication band A. Accordingly, coupling between the inductor **73**L and other high-frequency components is suppressed, being able to suppress degradation of a quality of a transmission signal in the communication band A and degradation of reception sensitivity in the communication band A.
(141) Further, for example, in the high-frequency module **1**A according to the present embodiment, the plurality of high-frequency components may include the duplexer **61** and the inductor **73**L, which is connected between the duplexer **61** and the antenna connection terminal **101**, and the inductor **73**L may be closer to the side surface **924** than the duplexer **61**.
(142) According to this configuration, the inductor **73**L can be arranged closer to the side surface **924**, which is not covered by the shield electrode layer **95**A, than the duplexer **61**. Since inductors are easily coupled with other high-frequency components in general, coupling between a plurality of high-frequency components via the shield electrode layer **95**A can be effectively suppressed. Further, the inductor **73**L is arranged on a transmission/reception path in the communication band A. Accordingly, coupling between the inductor **73**L and other high-frequency components is suppressed, being able to suppress degradation of a quality of a transmission signal in the communication band A and degradation of reception sensitivity in the communication band A.
(143) Further, for example, in the high-frequency module **1**A according to the present embodiment,

the plurality of high-frequency components may include the duplexer **61** and the inductor **73**L, which is connected between the duplexer **61** and the antenna connection terminal **101**, and the winding axis **73**La of a coil constituting the inductor **73**L may intersect with the side surface **924** but does not have to intersect with the side surface **921**.
(144) According to this configuration, the inductor **73**L can be arranged so that the winding axis **73**La of a coil constituting the inductor **73**L does not intersect with the side surface **921** covered by the shield electrode layer **95**A. In general, inductors generate a strong magnetic force along a winding axis of a coil and are prone to generate magnetic field coupling with components on the winding axis. Therefore, the configuration in which the winding axis **73**La does not intersect with the side surface **921** covered by the shield electrode layer **95**A can effectively suppress magnetic field coupling of the inductor **73**L with other high-frequency components via the shield electrode layer **95**A. Further, the inductor **73**L is arranged on a transmission/reception path in the communication band A. Accordingly, coupling between the inductor **73**L and other high-frequency components is suppressed, being able to suppress degradation of a quality of a transmission signal in the communication band A and degradation of reception sensitivity of the communication band A.
Third Embodiment
(145) A third embodiment will now be described. The present embodiment is mainly different from the above-described first embodiment in that three side surfaces of a resin member are covered by a shield electrode layer. The present embodiment will be described below focusing on the difference from the above-described first embodiment with reference to the accompanying drawing.
(146) The circuit configuration of a high-frequency module **1**B according to the present embodiment is the same as that of the high-frequency module **1** according to the first embodiment and the illustration and description thereof are therefore omitted.
(147) [3.1 Component Arrangement in High-Frequency Module **1**B]
(148) The component arrangement in the high-frequency module **1**B according to the present embodiment will now be specifically described with reference to FIG. **7**. FIG. **7** is a plan view of the high-frequency module **1**B according to the third embodiment.
(149) In the present embodiment, the side surface **921** of the resin member **92** corresponds to the first side surface and the side surfaces **922** to **924** correspond to the second side surface to the fourth side surface.
(150) A shield electrode layer **95**B covers the side surfaces **922** to **924** without necessarily covering the side surface **921** among the four side surfaces **921** to **924** of the resin member **92**. In other words, the shield electrode layer **95**B does not cover only one side surface **921** among the four side surfaces **921** to **924** of the resin member **92**.
(151) The inductor **71**L is an example of the first inductor and is arranged near the side surface **921** of the resin member **92**. The winding axis **71**La of a coil constituting the inductor **71**L is parallel to the side surfaces **922** and **924** and is orthogonal to the side surfaces **921** and **923**. That is, the winding axis **71**La intersects with the side surfaces **921** and **923**. Accordingly, the winding axis **71**La also intersects with the shield electrode layer **95**B covering the side surface **923**.
(152) A distance D**1** from an intersection P**1** between the winding axis **71**La and the side surface **921** to a center point C**1** of the inductor **71**L is shorter than a distance D**2** from an intersection P**2** between the winding axis **71**La and the side surface **923** to the center point C**1** of the inductor **71**L. That is, on the winding axis **71**La, the distance D**1** between the inductor **71**L and the side surface **921** is shorter than the distance D**2** between the inductor **71**L and the side surface **923**.
(153) The inductor **74**L is an example of a third inductor and is arranged near the center of the module board **91**. The winding axis **74**La of a coil constituting the inductor **74**L is parallel to the side surfaces **922** and **924** and is orthogonal to the side surfaces **921** and **923**. That is, the winding axis **74**La intersects with the side surfaces **921** and **923**. Accordingly, the winding axis **74**La also intersects with the shield electrode layer **95**B covering the side surface **923**.

(154) A distance D**3** from an intersection P**3** between the winding axis **74**La and the side surface **921** to a center point C**2** of the inductor **74**L is shorter than a distance D**4** from an intersection P**4** between the winding axis **74**La and the side surface **923** to the center point C**2** of the inductor **74**L. That is, on the winding axis **74**La, the distance D**3** between the inductor **74**L and the side surface **921** is shorter than the distance D**4** between the inductor **74**L and the side surface **923**.
(155) The inductor **75**L is an example of the third inductor and is arranged near the side surface **921** of the resin member **92**. The winding axis **75**La of a coil constituting the inductor **75**L is parallel to the side surfaces **922** and **924** and is orthogonal to the side surfaces **921** and **923**. That is, the winding axis **75**La intersects with the side surfaces **921** and **923**. Accordingly, the winding axis **75**La also intersects with the shield electrode layer **95**B covering the side surface **923**.
(156) A distance D**5** from an intersection P**5** between the winding axis **75**La and the side surface **921** to a center point C**3** of the inductor **75**L is shorter than a distance D**6** from an intersection P**6** between the winding axis **75**La and the side surface **923** to the center point C**3** of the inductor **75**L. That is, on the winding axis **75**La, the distance D**5** between the inductor **75**L and the side surface **921** is shorter than the distance D**6** between the inductor **75**L and the side surface **923**.
(157) [3.2 Method for Manufacturing High-Frequency Module **1**B]
(158) A method for manufacturing the high-frequency module **1**B will now be described with reference to FIG. **9**. FIG. **9** is a transparent view of a collective board **910**BC for high-frequency modules **1**B and **1**C according to the third and fourth embodiments.
(159) In the present embodiment, the collective board **910**BC including two module boards **91** is prepared, as illustrated in FIG. **9**. Then, a resin molding process, a film forming process, and a cutting process are executed with respect to the collective board **910**BC. Thus, the collective board **910**BC is cut after a shield electrode layer is formed on a surface of a resin member on the collective board **910**BC. On the module board **91** on the left side in the collective board **910**BC, the shield electrode layer **95**B is formed on three side surfaces **922** to **924** but the shield electrode layer **95**B is not formed on one side surface **921** among four side surfaces **921** to **924** of the resin member **92**. That is, the high-frequency module **1**B is manufactured.
(160) [3.3 Effects, Etc.]
(161) As described above, the high-frequency module **1**B according to the present embodiment includes: the module board **91** that has the main surfaces **91***a* and **91***b*, which are opposed to each other; a plurality of high-frequency components that are arranged on at least one of the main surfaces **91***a* and **91***b*; the resin member **92** that covers at least one of the main surfaces **91***a* and **91***b* and has a plurality of side surfaces, which include the side surfaces **921** to **924**, along the outer edge of the module board **91**; and the shield electrode layer **95**B that covers the side surfaces **922** to **924** without necessarily covering the side surface **921**.
(162) According to this configuration, the side surfaces **922** to **924** can be covered by the shield electrode layer **95**B without necessarily covering the side surface **921** by the shield electrode layer **95**B. A high-frequency component may be magnetically coupled, electrically coupled, or electromagnetically coupled with other high-frequency components via the shield electrode layer **95**B. Therefore, the configuration in which the side surface **921** among a plurality of side surfaces is not covered by the shield electrode layer **95**B can suppress such coupling and improve isolation between the plurality of high-frequency components. In addition, the shield electrode layer **95**B covers the side surfaces **922** to **924** among the plurality of side surfaces, being able to suppress external noise from entering the high-frequency components.
(163) Further, for example, in the high-frequency module **1**B according to the present embodiment, the plurality of high-frequency components may include the reception filter **61**R, the low noise amplifier **21**, and the inductor **71**L, which is connected between the reception filter **61**R and the low noise amplifier **21**, the winding axis **71**La of a coil constituting the inductor **71**L may intersect with each of the side surfaces **921** and **923**, and the distance D**1** from the intersection P**1** between the winding axis **71**La and the side surface **921** to the center point C**1** of the inductor **71**L may be

shorter than the distance D**2** from the intersection P**2** between the winding axis **71**La and the side surface **923** to the center point C**1** of the inductor **71**L.
(164) According to this configuration, when the winding axis **71**La of the inductor **71**L intersects with both of the side surface **921** which is not covered by the shield electrode layer **95**B and the side surface **923** which is covered by the shield electrode layer **95**B, the distance D**2** between the inductor **71**L and the side surface **923** on the winding axis **71**La can be set longer than the distance D**1** between the inductor **71**L and the side surface **921** on the winding axis **71**La. This can suppress coupling of the inductor **71**L with other high-frequency components via the shield electrode layer **95**B covering the side surface **923**.
(165) Further, for example, in the high-frequency module **1**B according to the present embodiment, the plurality of high-frequency components may include the transmission-reception filter **62** and the inductor **74**L, which is connected between the transmission-reception filter **62** and the antenna connection terminal **101**, the winding axis **74**La of a coil constituting the inductor **74**L may intersect with each of the side surfaces **921** and **923**, and the distance D**3** from the intersection P**3** between the winding axis **74**La and the side surface **921** to the center point C**2** of the inductor **74**L may be shorter than the distance D**4** from the intersection P**4** between the winding axis **74**La and the side surface **923** to the center point C**2** of the inductor **74**L.
(166) According to this configuration, when the winding axis **74**La of the inductor **74**L intersects with both of the side surface **921** which is not covered by the shield electrode layer **95**B and the side surface **923** which is covered by the shield electrode layer **95**B, the distance D**4** between the inductor **74**L and the side surface **923** on the winding axis **74**La can be set longer than the distance D**3** between the inductor **74**L and the side surface **921** on the winding axis **74**La. This can suppress coupling of the inductor **74**L with other high-frequency components via the shield electrode layer **95**B covering the side surface **923**.
(167) Further, for example, in the high-frequency module **1**B according to the present embodiment, the plurality of high-frequency components may include the transmission-reception filter **62** and the inductor **75**L, which is connected between the transmission-reception filter **62** and the antenna connection terminal **101**, the winding axis **75**La of a coil constituting the inductor **75**L may intersect with each of the side surfaces **921** and **923**, and the distance D**5** from the intersection P**5** between the winding axis **75**La and the side surface **921** to the center point C**3** of the inductor **75**L may be shorter than the distance D**6** from the intersection P**6** between the winding axis **75**La and the side surface **923** to the center point C**3** of the inductor **75**L.
(168) According to this configuration, when the winding axis **75**La of the inductor **75**L intersects with both of the side surface **921** which is not covered by the shield electrode layer **95**B and the side surface **923** which is covered by the shield electrode layer **95**B, the distance D**6** between the inductor **75**L and the side surface **923** on the winding axis **75**La can be set longer than the distance D**5** between the inductor **75**L and the side surface **921** on the winding axis **75**La. This can suppress coupling of the inductor **75**L with other high-frequency components via the shield electrode layer **95**B covering the side surface **923**.
(169) Further, for example, in the high-frequency module **1**B according to the present embodiment, the plurality of side surfaces may include the side surfaces **921** to **924** and the shield electrode layer **95**B may cover the side surfaces **922** to **924** without necessarily covering the side surface **921**.
(170) According to this configuration, three side surfaces **922** to **924** can be covered by the shield electrode layer **95**B. Accordingly, the high-frequency module **1**B can enhance suppression of external noise intrusion more than the high-frequency module **1** or **1**A according to the first or second embodiment.
Fourth Embodiment
(171) A fourth embodiment will now be described. The present embodiment is mainly different from the above-described first embodiment in that three side surfaces of a resin member are covered by a shield electrode layer. The present embodiment will be described below focusing on

the difference from the above-described first embodiment with reference to the accompanying drawing.
(172) The circuit configuration of a high-frequency module **1**C according to the present embodiment is the same as that of the high-frequency module **1** according to the first embodiment and the illustration and description thereof are therefore omitted.
(173) [4.1 Component Arrangement in High-Frequency Module **1**C]
(174) The component arrangement in the high-frequency module **1**C according to the present embodiment will now be specifically described with reference to FIG. **8**. FIG. **8** is a plan view of the high-frequency module **1**C according to the fourth embodiment.
(175) In the present embodiment, the side surface **923** of the resin member **92** corresponds to the first side surface and the side surfaces **921**, **922**, and **924** correspond to the second side surface to the fourth side surface.
(176) A shield electrode layer **95**C covers the side surfaces **921**, **922**, and **924** without necessarily covering the side surface **923** among the four side surfaces **921** to **924** of the resin member **92**. In other words, the shield electrode layer **95**C does not cover only one side surface **923** among the four side surfaces **921** to **924** of the resin member **92**.
(177) The inductor **72**L is an example of the second inductor and is arranged near the side surface **923** of the resin member **92**. The winding axis **72**La of a coil constituting the inductor **72**L is parallel to the side surfaces **922** and **924** and is orthogonal to the side surfaces **921** and **923**. That is, the winding axis **72**La intersects with the side surfaces **921** and **923**. Accordingly, the winding axis **72**La also intersects with the shield electrode layer **95**C covering the side surface **921**.
(178) A distance D**7** from an intersection P**7** between the winding axis **72**La and the side surface **923** to a center point C**4** of the inductor **72**L is shorter than a distance D**8** from an intersection P**8** between the winding axis **72**La and the side surface **921** to the center point C**4** of the inductor **72**L. That is, that is, on the winding axis **72**La, the distance D**7** between the inductor **72**L and the side surface **923** is shorter than the distance D**8** between the inductor **72**L and the side surface **921**.
(179) [4.2 Method for Manufacturing High-Frequency Module **1**C]
(180) A method for manufacturing the high-frequency module **1**C will now be described with reference to FIG. **9**.
(181) In the present embodiment, the collective board **910**BC including two module boards **91** is prepared, as illustrated in FIG. **9**. Then, a resin molding process, a film forming process, and a cutting process are executed with respect to the collective board **910**BC. Thus, the collective board **910**BC is cut after a shield electrode layer is formed on a surface of a resin member on the collective board **910**BC. On the module board **91** on the right side in the collective board **910**BC, the shield electrode layer **95**C is formed on three side surfaces **921**, **922**, and **924** but the shield electrode layer **95**C is not formed on one side surface **923** among four side surfaces **921** to **924** of the resin member **92**. The high-frequency module **1**C is thus manufactured.
(182) [4.3 Effects, Etc.]
(183) As described above, the high-frequency module **1**C according to the present embodiment includes: the module board **91** that has the main surfaces **91***a* and **91***b*, which are opposed to each other; a plurality of high-frequency components that are arranged on at least one of the main surfaces **91***a* and **91***b*; the resin member **92** that covers at least one of the main surfaces **91***a* and **91***b* and has a plurality of side surfaces, which include the side surfaces **921** to **924**, along the outer edge of the module board **91**; and the shield electrode layer **95**C that covers the side surfaces **921**, **922**, and **924** without necessarily covering the side surface **923**.
(184) According to this configuration, the side surfaces **921**, **922**, and **924** can be covered by the shield electrode layer **95**C without necessarily covering the side surface **923** by the shield electrode layer **95**C. A high-frequency component may be magnetically coupled, electrically coupled, or electromagnetically coupled with other high-frequency components via the shield electrode layer **95**C. Therefore, the configuration in which the side surface **923** among a plurality of side surfaces

is not covered by the shield electrode layer **95**C can suppress such coupling and improve isolation between the plurality of high-frequency components. In addition, the shield electrode layer **95**C covers the side surfaces **921**, **922**, and **924** among the plurality of side surfaces, being able to suppress external noise from entering the high-frequency components.
(185) Further, for example, in the high-frequency module **1**C according to the present embodiment, the plurality of high-frequency components may include the transmission filter **61**T, the power amplifier **11**, and the inductor **72**L, which is connected between the transmission filter **61**T and the power amplifier **11**, the winding axis **72**La of a coil constituting the inductor **72**L may intersect with each of the side surfaces **921** and **923**, and the distance D**7** from the intersection P**7** between the winding axis **72**La and the side surface **923** to the center point C**4** of the inductor **72**L may be shorter than the distance D**8** from the intersection P**8** between the winding axis **72**La and the side surface **921** to the center point C**4** of the inductor **72**L.
(186) According to this configuration, when the winding axis **72**La of the inductor **72**L intersects with both of the side surface **923** which is not covered by the shield electrode layer **95**C and the side surface **921** which is covered by the shield electrode layer **95**C, the distance D**8** between the inductor **72**L and the side surface **921** on the winding axis **72**La can be set longer than the distance D**7** between the inductor **72**L and the side surface **923** on the winding axis **72**La. This can suppress coupling of the inductor **72**L with other high-frequency components via the shield electrode layer **95**C covering the side surface **921**.
(187) Further, for example, in the high-frequency module **1**C according to the present embodiment, the plurality of side surfaces may include the side surfaces **921** to **924** and the shield electrode layer **95**C may cover the side surfaces **921**, **922**, and **924** without necessarily covering the side surface **923**.
(188) According to this configuration, three side surfaces **921**, **922**, and **924** can be covered by the shield electrode layer **95**C. Accordingly, the high-frequency module **1**C can enhance suppression of external noise intrusion more than the high-frequency module **1** or **1**A according to the first or second embodiment.
Other Embodiments
(189) The high-frequency modules and communication device according to the present disclosure have been described above based on the embodiments. However, the high-frequency modules and communication device according to the present disclosure are not limited to the above-described embodiments. The disclosure also includes other embodiments realized by combining any of the components in the above-described embodiments, modifications obtained by making various changes, which a person skilled in the art can think of, to the above-described embodiments without necessarily departing from the scope of the present disclosure, and various devices incorporating the above-described high-frequency modules and communication device.
(190) For example, two or three side surfaces among the four side surfaces of the resin member **92** are covered by the shield electrode layer in the above-described embodiments, but the configuration is not limited to this. For example, only one side surface may be covered by the shield electrode layer. An example of such a high-frequency module will be described with reference to FIG. **10**. FIG. **10** is a plan view of a high-frequency module **1**D according to another embodiment. In FIG. **10**, the side surface **921** of the resin member **92** corresponds to the first side surface, the side surface **922** corresponds to the second side surface, and the side surfaces **923** and **924** correspond to the third side surface and the fourth side surface. A shield electrode layer **95**D covers the side surface **922** without necessarily covering the side surfaces **921**, **923**, and **924** among the four side surfaces **921** to **924** of the resin member **92**. In other words, the shield electrode layer **95**D covers only one side surface **922** among the four side surfaces **921** to **924** of the resin member **92**. Thus, three side surfaces **921**, **923**, and **924** are not covered by the shield electrode layer **95**D in this configuration. Accordingly, the high-frequency module **1**D can enhance suppression of coupling between high-frequency components via the shield electrode layer **95** than the high-

frequency module according to each of the above-described embodiments.
(191) In each of the above-described embodiments, a plurality of high-frequency components is arranged on the main surface **91***a*, but the configuration is not limited to this. For example, the semiconductor integrated circuit **20** and/or the switch **51** may be arranged on the main surface **91***b*. That is, at least one of the plurality of high-frequency components may be arranged on the main surface **91***a* and at least one of the rest of the plurality of high-frequency components may be arranged on the main surface **91***b*. In this configuration, the main surface **91***b* may be also covered by the resin member as well as the main surface **91***a* and a post electrode and/or a bump electrode may be used as a plurality of external connection terminals **150**. In this configuration, components can be arranged on both surfaces of the module board **91**, being able to realize downsizing of the high-frequency module. Here, the switch **52** may be arranged on either the main surface **91***a* or **91***b*.
(192) Two switches **52** and **53** are used for connection and disconnection of the duplexer **61** and the transmission-reception filter **62** with respect to the power amplifier **11** and the low noise amplifier **21** in each of the above-described embodiments, but the configuration of the switches is not limited to this. For example, the switches **52** and **53** may be configured as a single switch. In this configuration, the single switch may include five terminals that are each connected with the transmission filter **61**T, the reception filter **61**R, the transmission-reception filter **62**, the output of the power amplifier **11**, and the input of the low noise amplifier **21**.
(193) The high-frequency module includes the switch **53** for switching filters connected to the low noise amplifier **21** in each of the above-described embodiments, but the high-frequency module does not have to include the switch **53**. In this configuration, the high-frequency module may include two low noise amplifiers instead of the low noise amplifier **21**. Here, the other end of the reception filter **61**R may be connected with one of the two low noise amplifiers and the other end of the transmission-reception filter **62** may be connected with the other of the two low noise amplifiers via the switch **52**.
(194) In the first and second embodiments described above, when two side surfaces are covered by the shield electrode layer, the two side surfaces are opposed to each other. However, the configuration is not limited to this. Two side surfaces covered by the shield electrode layer may intersect with each other. For example, the side surfaces **922** and **923** may be covered by the shield electrode layer **95** and the side surfaces **921** and **924** do not have to be covered by the shield electrode layer **95** in the first embodiment. Further, the side surfaces **923** and **924** may be covered by the shield electrode layer **95** and the side surfaces **921** and **922** do not have to be covered by the shield electrode layer **95**.
INDUSTRIAL APPLICABILITY
(195) The present disclosure can be widely used in communication devices such as mobile phones as a high-frequency module placed in a front-end section.
REFERENCE SIGNS LIST
(196) **1**, **1**A, **1**B, **1**C, **1**D high-frequency module **2** antenna **3** RFIC **4** BBIC **5** communication device **11** power amplifier **20** semiconductor integrated circuit **21** low noise amplifier **51**, **52**, **53** switch **61** duplexer **61**R reception filter **61**T transmission filter **62** transmission-reception filter **71**, **72**, **73**, **74**, **75** matching circuit **71**L, **72**L, **73**L, **74**L, **75**L inductor **71**La, **72**La, **73**La, **74**La, **75**La winding axis **91** module board **91***a*, **91***b* main surface **92** resin member **95**, **95**A, **95**B, **95**C, **95**D shield electrode layer **101** antenna connection terminal **111** high-frequency input terminal **121** high-frequency output terminal **150** external connection terminal **511**, **512**, **513**, **521**, **522**, **523**, **524**, **531**, **532**, **533** terminal **910**, **910**BC collective board **921**, **922**, **923**, **924** side surface **925** top surface C**1**, C**2**, C**3**, C**4** center point D**1**, D**2**, D**3**, D**4**, D**5**, D**6**, D**7**, D**8** distance P**1**, P**2**, P**3**, P**4**, P**5**, P**6**, P**7**, P**8** intersection

Claims

1. A high-frequency module comprising: a module board that comprises a first main surface and a second main surface, the first main surface and the second main surface being opposed to each other; a plurality of high-frequency components that are on at least one of the first main surface and the second main surface; a resin member that covers the at least one of the first main surface and the second main surface and comprises a plurality of side surfaces, the plurality of side surfaces comprising a first side surface and a second side surface, along an outer edge of the module board; and a shield electrode layer that covers the second side surface without covering the first side surface, wherein the plurality of high-frequency components comprise: a first filter, a low noise amplifier, a first inductor that is connected between the first filter and the low noise amplifier, and a first winding axis of a coil constituting the first inductor intersects with the first side surface but does not intersect with the second side surface.
2. The high-frequency module according to claim 1, wherein the first inductor is closer to the first side surface than the second side surface.
3. The high-frequency module according to claim 2, wherein the plurality of high-frequency components further comprise: a second filter, a power amplifier, and a second inductor that is connected between the second filter and the power amplifier, wherein the second inductor is closer to the first side surface than the second side surface.
4. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a first filter, a low noise amplifier, and a first inductor that is connected between the first filter and the low noise amplifier, wherein the first inductor is closer to the first side surface than the low noise amplifier.
5. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a first filter, a low noise amplifier, a first inductor that is connected between the first filter and the low noise amplifier, and the first winding axis of a coil constituting the first inductor intersects with each of the first side surface and the second side surface, wherein a first distance from a first intersection between the first winding axis and the first side surface to a center point of the first inductor is shorter than a second distance from a second intersection between the first winding axis and the second side surface to the center point of the first inductor.
6. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a second filter, a power amplifier, and a second inductor that is connected between the second filter and the power amplifier, wherein the second inductor is closer to the first side surface than the second side surface.
7. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a second filter, a power amplifier, and a second inductor that is connected between the second filter and the power amplifier, wherein the second inductor is closer to the first side surface than the power amplifier.
8. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a second filter, a power amplifier, a second inductor that is connected between the second filter and the power amplifier, and a second winding axis of a coil constituting the second inductor intersects with the first side surface but does not intersect with the second side surface.
9. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a second filter, a power amplifier, a second inductor that is connected between the second filter and the power amplifier, a second winding axis of a coil constituting the second inductor intersects with each of the first side surface and the second side surface, wherein a first distance from a first intersection between the second winding axis and the first side surface to a center point of the second inductor is shorter than a second distance from a second intersection

between the second winding axis and the second side surface to the center point of the second inductor.

10. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a third filter, and a third inductor that is connected between the third filter and an antenna connection terminal, wherein the third inductor is closer to the first side surface than the second side surface.

11. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a third filter, and a third inductor that is connected between the third filter and an antenna connection terminal, wherein the third inductor is closer to the first side surface than the third filter.

12. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a third filter, a third inductor that is connected between the third filter and an antenna connection terminal, and a third winding axis of a coil constituting the third inductor intersects with the first side surface but does not intersect with the second side surface.

13. The high-frequency module according to claim 1, wherein the plurality of high-frequency components comprise: a third filter, a third inductor that is connected between the third filter and an antenna connection terminal, and the third winding axis of a coil constituting the third inductor intersects with each of the first side surface and the second side surface, and a first distance from a first intersection between the third winding axis and the first side surface to a center point of the third inductor is shorter than a second distance from a second intersection between the third winding axis and the second side surface to the center point of the third inductor.

14. The high-frequency module according to claim 1, wherein the plurality of side surfaces further comprise a third side surface and a fourth side surface, and the shield electrode layer covers the second side surface and the fourth side surface without covering the first side surface and the third side surface.

15. The high-frequency module according to claim 14, wherein the first side surface and the third side surface are opposed to each other, and the second side surface and the fourth side surface are opposed to each other.

16. The high-frequency module according to claim 1, wherein the plurality of side surfaces further comprise a third side surface and a fourth side surface, and the shield electrode layer covers the second side surface, the third side surface, and the fourth side surface without covering the first side surface.

17. The high-frequency module according to claim 1, wherein the plurality of side surfaces further comprise a third side surface and a fourth side surface, and the shield electrode layer covers the second side surface without covering the first side surface, the third side surface, and the fourth side surface.

18. A communication device comprising: a signal processing circuit that is configured to process a high-frequency signal; and the high-frequency module according to claim 1 that is configured to transmit the high-frequency signal between the signal processing circuit and an antenna.